

How Modern Programmers Think

Course Outline

About This Course

This course provides aspiring programmers with a solid foundation in core programming concepts, system navigation, file management, and data handling. Alongside these technical skills, it also develops essential soft skills and productivity habits. Designed to build a well-rounded skill set, the program prepares learners for the demands of modern software development through hands-on practice in Windows and other operating systems. By connecting theory with real-world application, the curriculum sets the stage for further learning in areas like web development, backend programming, and data analysis.

- **Duration:** Five days (~35 instructional hours).
- **Delivery:** Instructor-led, with live coding demonstrations, hands-on activities, and reference documentation.
- **Audience:** People new to computer programming — aspiring programmers building a foundation before moving into specialized, hands-on courses.
- **Prerequisites:** Comfort working with a computer — navigating the file system, editing files, and similar tasks. No prior programming experience is assumed.
- **Platform:** Windows-first, with an overview of macOS and Linux; terminal examples use PowerShell.

What You Will Learn

Equip participants with the essential foundation needed to learn modern programming disciplines such as front-end web development, back-end programming, and data analysis.

- **Core Programming Skills:** Strengthen understanding of computational thinking, algorithms, and logical problem-solving.
- **System Proficiency:** Develop confidence in navigating and managing files across operating systems, with a focus on Windows environments for enhanced productivity.

- **Introduction to Data Concepts:** Build a foundational understanding of databases, spreadsheets, and APIs to grasp data manipulation and storage in software development.
 - **Resource Optimization:** Improve problem-solving through effective use of online learning resources and documentation.
 - **Well-being and Productivity:** Learn strategies to maintain balance and focus, supporting long-term engagement and success in coding.
 - **Collaboration and Code Analysis:** Understand teamwork's role in programming and learn to assess code quality, readability, and technical debt through real examples.
 - **Applying Programming Foundations:** Explore real-world applications of programming principles through practical code analysis and hands-on exercises.
 - **Future Preparedness:** Establish a strong base for advancing into hands-on programming courses in web development, Java, and related fields.
-

Module I — Introduction to Modern Programming

- A. **Defining the Modern Programmer:** Understanding today's programming landscape, tools, and expectations.
- B. **Who This Course Is For:** Learners transitioning into tech or strengthening foundational skills for web, backend, or data development.

Module II — Productivity, Focus, and Developer Well-Being

- A. **Organizing Your Workflow:** Building habits that improve clarity, structure, and time management.
- B. **Sharpening Focus and Attention to Detail:** Techniques to boost productivity and accuracy in coding.
- C. **Staying Healthy in Tech:** Strategies to prevent burnout, manage screen time, and sustain long-term growth.

Module III — Development Environments & Technical Efficiency

- A. **Operating Systems Overview:** Understanding the role of Windows, macOS, and Linux in software development.

- B. **Windows Essentials:** Command-line tools, environment variables, and basic scripting.
- C. **Text Editors and IDEs:** Mastering VS Code and shortcuts for faster development.
- D. **Smart Research Practices:** Efficient use of browsers, search techniques, and personal organization tools.
- E. **Leveraging Online Coding Resources:** Finding reliable documentation, tutorials, and communities.

Module IV — Software Development Lifecycle (SDLC) & Collaboration

- A. **The Big Picture:** Key phases, roles, and success factors in software projects.
- B. **Working Locally with Version Control:** Managing changes and maintaining clean workflows.
- C. **Hands-On with Git:** Cloning course demos, creating branches, and contributing code.
- D. **Writing in Markdown:** Documenting projects clearly and professionally.
- E. **Collaboration & Code Quality:** Understanding technical debt, teamwork, and review best practices.

Module V — Computational Thinking, Data Structures & Algorithms

- A. **Problem-Solving with Computational Thinking:** Breaking down problems into logical steps.
- B. **Core Data Structures:** Lists, arrays, and objects for organizing information.
- C. **Control Flow Mastery:** Writing efficient logic with conditionals and loops.
- D. **Algorithm Basics:** Comparing linear and binary search to understand performance tradeoffs.

Module VI — Data Management: From Spreadsheets to Databases

- A. **Foundations of Data Systems:** How data is structured, stored, and retrieved.
- B. **Practical Tools:** Using CSVs and spreadsheets for basic analysis.

- C. **Introduction to Databases:** Exploring SQL and NoSQL systems and their use in modern applications.

Module VII — Programming Syntax & Logic

- A. **Anatomy of a Programming Language:** Understanding syntax, semantics, and execution.
- B. **JavaScript Fundamentals:** Exploring a flexible, interpreted language for web and data applications.
- C. **Hands-On Practice in VS Code:** Writing and debugging simple programs.
- D. **Modern Web Development Overview:** How front-end and back-end technologies work together.
- E. **Introduction to Java:** Learning the principles of a strongly typed, compiled language.

Module VIII — Building for the Future

- A. **Bridging Theory to Practice:** Applying foundational skills to real-world projects.
- B. **Next Steps:** Preparing for specialized learning in web development, Java, and data analysis.